

JUANDA SURABAYA, Indonesia

WMO: 969350

Lat: 7.3747S

Lon: 112.7820E

Elev: 3

StdP: 101.29

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.2	22.1	16.6	11.8	29.8	17.6	12.6	29.3	8.7	29.9	8.2	29.7	0.4	270	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	8.3	34.1	24.7	33.5	24.8	33.1	25.0	27.1	31.2	26.8	30.9	26.6	30.6	4.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.5	29.1	25.7	21.0	28.9	25.5	20.8	28.8	85.4	31.6	84.3	31.2	83.4	30.9	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.1	6.2	DB	20.1	35.1	1.0	0.7	19.4	35.6	18.8	36.0	18.2	36.4	17.5	36.9	(4)
(5)			WB	18.2	28.2	1.6	0.9	17.1	28.8	16.1	29.4	15.3	29.8	14.1	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	28.1	28.0	28.2	28.5	28.4	27.8	27.1	27.0	27.8	29.0	29.1	28.5	(6)	
	DBStd	1.10	0.91	0.79	0.78	0.79	0.90	0.94	0.96	0.99	0.93	0.82	1.01	0.99	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6612	560	503	564	555	570	533	531	527	533	588	573	573	(10)	
	CDD18.3	3570	302	270	306	305	312	283	273	269	283	330	323	314	(11)	
	CDH23.3	39851	2952	2620	3182	3554	3748	3228	2942	2920	3289	4137	3920	3359	(12)	
(13)	CDH26.7	15697	1014	877	1140	1407	1518	1218	1052	1097	1411	1927	1760	1276	(13)	
(14)	Wind	WSAvg	2.5	2.6	2.7	2.1	2.1	2.4	2.6	2.7	3.0	2.8	2.2	2.1	(14)	
(15) Precipitation	PrecAvg	1650	314	285	294	187	120	55	28	8	9	29	112	208	(15)	
	PrecMax	2609	571	454	496	297	258	253	124	89	111	123	294	417	(16)	
	PrecMin	1097	128	128	183	73	12	0	0	0	0	0	0	44	(17)	
	PrecStd	422	107	97	93	76	84	69	38	23	28	40	98	96	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	33.1	33.2	33.1	33.0	32.2	32.1	32.2	33.5	34.6	35.0	34.2	(19)	
		MCWB	26.0	25.9	26.1	25.9	25.8	25.1	24.4	24.1	23.0	23.7	24.7	26.0	(20)	
	2%	DB	32.7	32.3	32.5	32.5	32.3	31.9	31.4	31.7	32.9	34.0	34.2	33.2	(21)	
		MCWB	26.1	25.9	26.0	25.9	25.6	25.1	24.2	23.8	23.4	24.0	24.9	25.8	(22)	
	5%	DB	31.9	31.8	31.9	32.1	32.0	31.3	31.0	31.2	32.3	33.4	33.6	32.6	(23)	
		MCWB	25.9	25.9	26.0	25.9	25.6	24.9	24.1	23.6	23.5	24.1	25.1	25.8	(24)	
	10%	DB	31.1	31.0	31.2	31.6	31.5	31.0	30.6	30.8	31.9	32.9	32.8	31.8	(25)	
		MCWB	25.8	25.8	25.9	25.9	25.4	24.8	24.0	23.5	23.5	24.2	25.4	25.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.1	27.1	27.3	27.5	27.2	26.7	26.5	26.1	26.2	26.6	27.0	27.2	(27)	
		MCDB	31.7	31.3	31.4	31.2	31.0	30.3	30.3	29.9	30.1	31.0	31.6	32.1	(28)	
	2%	WB	26.7	26.6	26.8	27.0	26.8	26.4	26.0	25.5	25.6	26.2	26.6	26.7	(29)	
		MCDB	31.0	30.7	30.7	30.7	30.6	29.9	29.4	29.5	29.8	30.6	31.2	31.4	(30)	
	5%	WB	26.3	26.2	26.5	26.7	26.6	26.1	25.5	25.0	25.1	25.8	26.3	26.5	(31)	
		MCDB	30.5	30.2	30.3	30.3	30.2	29.6	29.0	29.3	29.5	30.3	30.8	30.9	(32)	
	10%	WB	26.0	26.0	26.2	26.4	26.2	25.8	25.1	24.5	24.7	25.5	26.1	26.1	(33)	
		MCDB	30.1	29.9	29.9	30.0	29.8	29.2	28.6	28.9	29.3	30.1	30.6	30.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	6.8	6.8	6.4	6.4	6.7	7.4	8.1	8.6	8.3	7.6	7.1	(35)	
		MCDBR	7.7	7.7	7.4	6.9	6.8	7.0	7.7	8.2	9.0	8.9	8.5	8.0	(36)	
		MCWBR	2.7	2.7	2.6	2.3	2.2	2.4	2.6	2.7	2.7	2.6	2.5	2.6	(37)	
	5% WB	MCDBR	7.2	7.1	6.8	6.4	6.0	5.9	6.4	7.0	7.7	7.8	7.5	7.3	(38)	
		MCWBR	2.7	2.7	2.7	2.5	2.2	2.3	2.4	2.5	2.5	2.5	2.6	2.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.534	0.535	0.527	0.509	0.478	0.453	0.440	0.458	0.475	0.544	0.583	0.568	(40)	
	taud		2.097	2.098	2.122	2.166	2.253	2.324	2.354	2.277	2.219	2.020	1.932	1.982	(41)	
	Ebn at Noon		819	817	811	798	795	801	821	830	842	801	773	787	(42)	
	Edh at Noon		171	171	165	151	132	120	118	133	147	183	200	191	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.90	5.00	5.11	5.11	5.00	4.90	5.20	5.82	6.33	6.12	5.43	4.81	(44)	
	RadStd		0.44	0.34	0.33	0.34	0.35	0.34	0.29	0.19	0.33	0.55	0.48	0.40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.24	+0.73	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.24	+0.73	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page